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Paper 1: The Evolving Role of In Vitro–In Vivo Correlation in Model-Informed Drug Develop

Bibliographic Information

- **Title:** The Evolving Role of In Vitro–In Vivo Correlation in Model-Informed Drug Development: A Multi-Stakeholder Perspective
- **Authors:** Chenel M, et al.
- **Journal:** CPT: Pharmacometrics & Systems Pharmacology, 2025
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Paper Type: GENERIC

Quick Take

This multi-stakeholder perspective reviews the current and future role of IVIVC/R and PBBM in drug development, highlighting opportunities to reduce in vivo studies and accelerate regulatory decisions, while identifying key gaps in regulatory guidance, cross-functional collaboration, and integration of new technologies.

Executive Summary

This multi-stakeholder perspective paper consolidates views from academia, industry, service providers, and regulators on the evolving role of in vitro–in vivo correlation/relationship (IVIVC/R) and physiologically based biopharmaceutics modeling (PBBM) in drug development. It highlights opportunities to reduce costly in vivo studies, accelerate development, and support regulatory decisions, while also identifying key challenges: the need for better understanding of colonic absorption, integration of new technologies (organoids, AI/ML), clearer regulatory guidelines for PBBM, and improved cross-functional collaboration. The paper advocates for a gradual merger of mechanistic PBBM and data-driven PopPK approaches to leverage the strengths of both.

Scientific Context & Motivation

In vitro–in vivo correlation/relationship (IVIVC/R) models, including physiologically based biopharmaceutics modeling (PBBM), are essential tools for linking drug product quality attributes to clinical performance. They can reduce the need for costly in vivo bioequivalence studies and support regulatory decisions. However, their broader adoption is hindered by a lack of clear regulatory guidelines (especially for PBBM), challenges in model validation, and the need for better integration of mechanistic and data-driven approaches. This paper addresses these issues from multiple stakeholder perspectives.

✂ Methodological Snapshot

This is a perspective/review article, not a primary research study. It synthesizes expert opinions and published literature to provide a multi-stakeholder view on IVIVC/R and PBBM. The methodology is qualitative and descriptive, drawing on the collective experience of the authors from academia, industry, service providers, and regulatory science. No new data or models are presented. The paper references established frameworks (BCS, FDA/EMA guidelines, ICH M15) and discusses their application and limitations.

📋 Detailed Analysis

The paper provides a comprehensive, multi-stakeholder perspective on the current and future role of IVIVC/R and PBBM in model-informed drug development. It effectively contrasts the strengths of mechanistic PBBM (better extrapolation, mechanistic insight) with data-driven PopPK (variability estimation, covariate effects) and highlights the emerging convergence of these approaches. The discussion of regulatory gaps—particularly the lack of dedicated PBBM guidance in Europe and the absence of consensus for generic drugs—is timely and actionable. The paper also identifies critical competency gaps (e.g., need for consultants with deep biopharmaceutics expertise) and organizational challenges (cross-functional alignment, data workflows). However, the paper is primarily descriptive and lacks quantitative examples or case studies that would illustrate the practical application of the concepts discussed. The supplementary materials may contain such examples, but they are not analyzed here. The call for clearer data requirements and model validation standards for PBBM is well-founded and echoes ongoing discussions in the field.

Domain Context

This paper sits at the intersection of pharmaceutical sciences, regulatory science, and pharmacometrics. It addresses the evolving landscape of biopharmaceutics modeling, where traditional IVIVC (based on deconvolution/convolution) is being supplemented or replaced by mechanistic PBBM. The multi-stakeholder format (academia, industry, service providers, regulators) is valuable for identifying shared challenges and divergent perspectives. The paper is particularly relevant given the recent ICH M15 draft guidance on MIDD and ongoing FDA/EMA initiatives for model-integrated evidence in generic drugs.

📌 Key Findings

1. IVIVC/R and PBBM are critical for linking biopharmaceutical properties to clinical performance, reducing in vivo studies, and supporting regulatory decisions.
2. PBBM is increasingly used to complement or replace traditional IVIVC, especially for complex formulations and early development.
3. A key gap is the lack of dedicated regulatory guidance for PBBM, particularly in Europe and for generic drugs.
4. Emerging technologies (organoids, AI/ML) and hybrid modeling approaches (PBBM + PopPK) offer promising avenues for improving predictive accuracy.
5. Successful implementation requires cross-functional alignment (DMPK, CMC, Clinical Pharmacology, Regulatory) and robust internal processes.

Strengths & Limitations

Strengths

- Comprehensive multi-stakeholder perspective covering academia, industry, service providers, and regulators.
- Timely discussion of regulatory gaps and ongoing initiatives (ICH M15, FDA pilot program).
- Clear articulation of the complementary strengths of PBBM and PopPK and the potential for convergence.
- Identification of practical organizational and competency gaps that hinder adoption.

Limitations (Acknowledged by Authors)

- Lack of dedicated regulatory guidance for PBBM in Europe.
- Insufficient model validation and inadequate in vivo PK data as barriers to PBBM biowaivers.
- Need for better understanding of colonic absorption and new in vivo reference data.
- Challenges in integrating formulation properties into models.

Limitations (Expert Review)

- The paper is primarily descriptive and lacks quantitative case studies or data to support its claims.
- The multi-stakeholder perspective may not fully represent all viewpoints (e.g., patient advocacy, payers).
- The discussion of emerging technologies (organoids, AI/ML) is superficial and lacks specific examples or validation data.
- The paper does not address the cost or resource implications of implementing advanced modeling approaches.

Generalizability

The perspectives are broadly applicable to oral drug development, with some extension to other routes (long-acting injectables, transdermal). The regulatory discussion is focused on FDA and EMA, limiting generalizability to other regions.

Figures & Tables

- **Figure 1:** Comparison of PBBM and PopPK approaches in an IVIVC/IVIVR setting, highlighting their respective advantages and the envisioned future merger.
 - *Significance:* Central to the paper's thesis: illustrates the complementary strengths of mechanistic and data-driven approaches and the expected convergence.

Code & Reproducibility Assessment

Not applicable (perspective/review article). No code or data analysis presented.

Supplementary Materials

The supplementary material (psp470137-sup-0001-supinfo.docx) likely contains regulatory guideline references and case examples of IVIVC/R applications, but its content was not analyzed in detail here.

Future Directions

Future work should focus on: (1) developing and validating hybrid PBBM-PopPK models that integrate mechanistic insight with variability estimation; (2) establishing clear regulatory guidelines for PBBM data requirements and validation, particularly for generic drugs; (3) leveraging emerging technologies (organoids, AI/ML, molecular dynamics) to improve in vitro-in vivo translation; (4) conducting prospective case studies that demonstrate the impact of these models on regulatory decisions and development timelines.

Expert Commentary

As a senior pharmacometrician, I find this paper a valuable synthesis of current thinking in biopharmaceutics modeling. The emphasis on ‘fit-for-purpose’ modeling and the need to move beyond black-box solutions resonates with best practices in model-informed drug development. The identified gap in regulatory guidance for PBBM, especially for generic drugs, is a critical barrier that the field must address. The proposed merger of PBBM and PopPK is conceptually appealing but will require significant methodological development and validation. The paper would have benefited from more concrete examples or case studies to illustrate the practical challenges and solutions. Nonetheless, it serves as an excellent resource for teams looking to align on strategy and identify areas for investment.

Bottom Line

IVIVC/R and PBBM are powerful tools for reducing in vivo studies and accelerating drug development, but their broader adoption requires clearer regulatory guidelines, better integration of mechanistic and data-driven approaches, and enhanced cross-functional collaboration. Practitioners should invest in fit-for-purpose modeling, early regulatory engagement, and robust internal processes to maximize impact.

☒ Figures

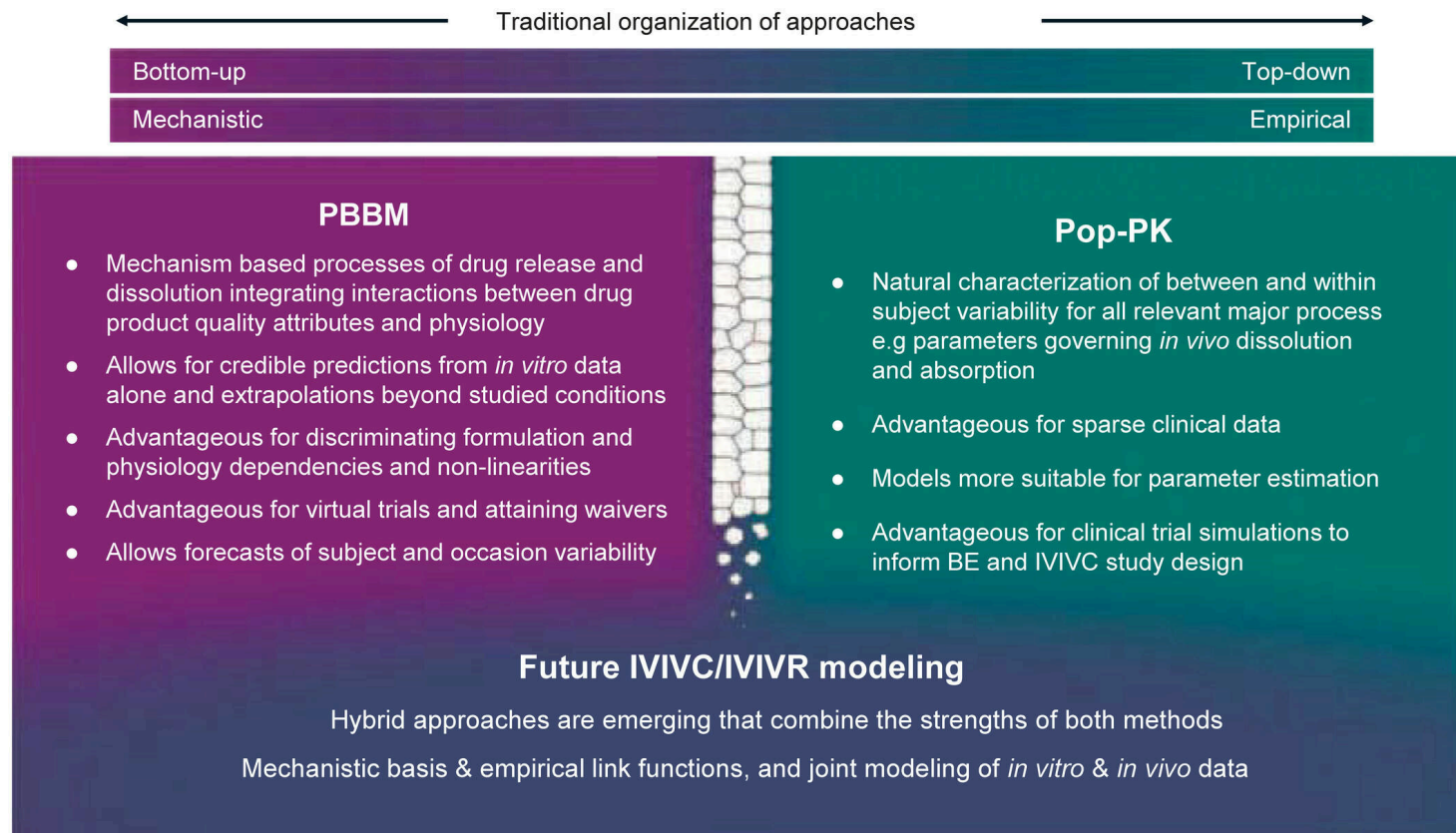


Figure 1: The PBBM and PopPK approaches have different advantages and current use in an IVIVC/IVIVR setting. In the future we foresee a gradual merger of these approaches